Think like a Bayesian **ANSWER KEY**

Restaurant stars and well water

Work in pairs. Keep a phone calculator handy.

One idea for today

Combine what you **knew before** (the **prior**) with what the **data** say. The more data you have, the more you trust them. The result is your **best guess** (the **posterior**).

Exercise 1. Which restaurant?

8 minutes

Recently, **10 people** tried restaurants in the area. Their average rating was **4.0 stars**. That is our **prior**: what a typical restaurant here gets. We **borrow** these 10 ratings and add them to each restaurant's own reviews. The best guess is then simply the **average of all the ratings**, borrowed and own:

$$\text{best guess} = \frac{\text{total stars}}{\text{total ratings}} = \frac{10 \times 4.0 \ + \ \text{own reviews} \times \text{own average}}{10 \ + \ \text{own reviews}}$$

1. Restaurant A has 2 reviews, average 5.0. Restaurant B has 800 reviews, average 4.7. Which one would you pick? ☐ A ■ B *(most people pick B)*

2. Fill in the table. Restaurant A is done for you.

Restaurant	Own reviews	Own average	Total stars 40 + reviews × average	Total ratings	Best guess
A	2	5.0	40 + 2 × 5.0 = 50	12	50/12 = 4.17
B	800	4.7	**3,800**	**810**	**4.69**
D	1	1.0	**41**	**11**	**3.73**

3. What share of each restaurant's ratings are its own? A: 2/12 = **0.17** B: 800/810 = **0.99**

This share is how much we **trust its own reviews**. Why did A drop from 5.0 to 4.17 while B hardly moved? *Only 17% of A's ratings are its own: the 10 borrowed ratings dominate. For B, 99% are its own, so the borrowed ratings hardly count.*

4. D has one 1-star review, yet its best guess is 3.73. Why? *Its one review is swamped by the 10 borrowed ones: with little data, the guess stays near 4.0, whether the data are high or low.*

Exercise 2. Is the well water safe? *10 minutes*

A family tests its well for arsenic **ten times**. The EPA limit is **10 μg/L**. Question: **how often is the water above the limit?**

Prior: in nearby wells, 5 of 10 past tests were above the limit.

> **Keep score.** Start with **10 tests borrowed** from the nearby wells: **Above = 5**, **Below = 5**. After each of the family's tests, add 1 to Above or to Below.
>
> **Best guess** = $\dfrac{\text{Above}}{\text{Above + Below}}$ (2 decimals)

5. Fill in the score card. The first rows are done for you.

Test	Arsenic (μg/L)	Above or below 10?	Above	Below	Best guess
—	*nearby wells (prior)*		5	5	0.50
1	14	above	6	5	0.55
2	7	below	6	6	0.50
3	12	**above**	**7**	**6**	**0.54**
4	18	**above**	**8**	**6**	**0.57**
5	11	**above**	**9**	**6**	**0.60**
6	15	**above**	**10**	**6**	**0.63**
7	9	**below**	**10**	**7**	**0.59**
8	13	**above**	**11**	**7**	**0.61**
9	14	**above**	**12**	**7**	**0.63**
10	14	**above**	**13**	**7**	**0.65**

6. The well's own tests alone: **8** of 10 above, a share of **0.80**. The nearby wells: 0.50. Your final best guess: **0.65**.

Where does it fall? **halfway between the two**

7. Why halfway? The final score card holds 20 tests: 10 borrowed and **10** of the family's own. Both count equally, so

$$0.5 \times 0.80 \;+\; 0.5 \times 0.50 \;=\; \mathbf{0.65}$$

8. A second family tested only **twice**, both above the limit. Their score: Above = **7**, Below = **5**, best guess = **0.58**.

Which family's guess stays closer to the neighbours' 0.50? **the second family** Why? *They have only 2 tests of their own, so the prior weighs more.*

Exit ticket *2 minutes*

Sullivan County expects 7 lung cancer cases; Philadelphia expects 1,219. Both SIRs are pulled toward the state average. Which one is pulled more?

■ Sullivan ☐ Philadelphia *Like Restaurant A and the second family: little data, so the prior weighs more.*